Thermal Imaging-based Edge Perception for Human Transit Surveillance Applications

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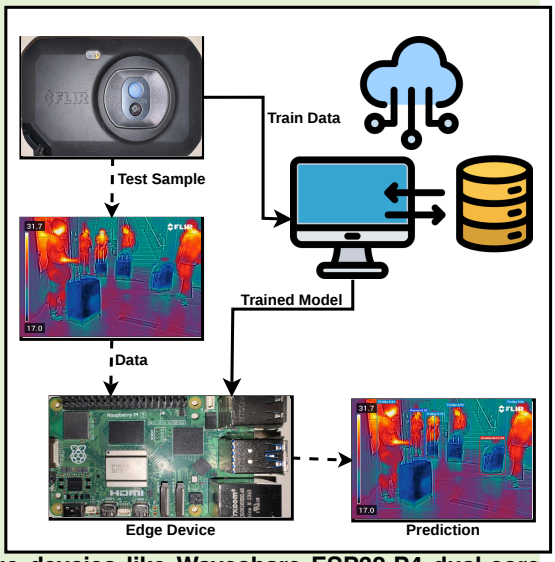
***Abstract*— Automated surveillance in high-throughput transit hubs, such as airport terminals, railway stations, and multi-modal transit corridors, requires robust round the clock perception while strictly preserving personal privacy. Long-wave infrared (LWIR) thermal sensors provide illumination-invariant sensing and inherent privacy preservation by imaging radiometric temperature differentials rather than identifying visual textures. In this work, we curated the thermal transit surveillance data for identifying humans, humans with trolley, humans with bag and stationary safety hazards like unattended anomaly using Teledyne FLIR C5 thermal camera. On the curated dataset of 1000+ thermal images with 3,849 annotated instances, we developed a new lightweight thermal Hybrid Dilated ECA-YOLO architecture and compared its performance with the existing YOLO state-of-the-art. Compared to existing literature, the proposed Hybrid Dilated ECA-YOLO embeds dilated residual bottlenecks in deeper feature stages and couples with parameter-free 1D Efficient Channel Attention (ECA) to capture long-range contextual spatial relationships. The proposed model achieves 87.89% mAP@50 on the held-out test split with only 3.01 M parameters and 8.1 GFLOPs. Crucially, it yields massive recall improvements on security-critical targets (+8.5pp for Person, +10.2pp for Bag, and +11.0pp for Unattended Anomaly) and out-localizes larger 11 M+ parameter models such as YOLOv8s and YOLO11s. Finally, the proposed model is hardware deployable and inferecned on various edge devcies like Waveshare ESP32-P4 dual-core RISC-V microcontroller, Raspberry Pi 5 SBC, and an NVIDIA Jetson AGX Xavier. An in-line physical energy profiling is conduceted using Joulescope JS220 high-speed DC energy analyzer (2 MS/s) across the edge devices and achived inference (energy efficiency) of xx (yy), xx (yy), and xx (yy) for ESP32-P4, Raspberry Pi 5 SBC, NVIDIA Jetson AGX Xavier, respectively. These results showcase, the proposed end-to-end pipelined product as a practical, energy-efficient solution for autonomous privacy-preserving transit security.**



***Index Terms*— Object detection, Thermal infrared sensing, Transit hubs, Unattended luggage anomaly, YOLO.**

I. Introduction

Infrared (IR) and thermal imaging sensors have become fundamental components of modern perception systems under poor lighting and adverse weather conditions [1]. Unlike visible-light cameras, thermal sensors operate independently of ambient illumination in Long-Wave Infrared (LWIR) spectral band (7.5–13.5 μm) and inherently preserve privacy by capturing temperature-dependent information rather than identifiable textures. These properties make them suitable for intelligent transportation [2], [3], industrial monitoring [4], [5], assistive mobility [6], [7], smart-home [8], pedestrian detection [9], [10] , indoor occupancy monitoring [11] , and target tracking [12], and smart-city monitoring applications [13]. The integration of thermal sensing with embedded AI has further enabled intelligent edge nodes capable of local decision-making and event recognition without continuous cloud connectivity, thereby reducing latency, communication overhead, and energy consumption [3], [7], [8].

Commercial systems including FLIR Tau, A655Sc, and ThermiBot 2 have demonstrated effectiveness for pedestrian detection and traffic analytics [10], [14], [15]. Despite these advances, most IR-based perception frameworks treat all pedestrians as a homogeneous class, overlooking the diversity of human mobility and their relation with surrounding objects in real environments. For example, individuals may walk unassisted (free walk), carry objects such as bags and trollys during transit. Most common transit-hubs are international airport terminals, railway concourses, and metro transit stations. The transit allows high-throughput with various carry objects and requires ensuring public safety, crowd flow optimization, and anomaly detection. Even though the visible cameras can capture identifiable facial traits and carry object, the optical-based solution raises severe privacy and data protection con-

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cerns under strict regulatory frameworks such as the European General Data Protection Regulation (GDPR) [6], [13]. While early works addressed generic thermal pedestrian detection [3], [10], [15], and recent studies explored assistive mobility detection [7], [16], these efforts mainly focus on binary recognition or human-carry object. Furthermore, the computational complexity of current DL frameworks poses challenges for deployment on resource-constrained edge devices.

According to Airports Council International (ACI) World and the SITA Baggage IT Insights report, 33.4 million bags mishandled, delayed, or misplaced across international transit hubs annually [17], [18]. Simultaneously, public transit concourses and landside airport terminals remain prime targets for improvised explosive devices (IEDs) concealed within abandoned personal luggage [19]. Because international aviation security protocols (e.g., ICAO Annex 17 and TSA guidelines) mandate that any unattended bag be treated as a potential explosive threat until cleared by explosive ordnance disposal (EOD) teams, even benign forgotten bags routinely trigger terminal evacuations, ground stops, and cascading economic losses reaching hundreds of thousands of dollars per incident [20]. On the otherhand, optical sensors suffer low-light night conditions, heavy artificial glare, abrupt shadows, backlighting through glass facades, and atmospheric obscuration. Also, the usage of optical raises to the privacy preserving aspect to travellers. Contraryly, long-wave radiation is privacy persevering, but inherently imaging is blurry and lacks high-frequency color or texture edges. In transit hubs, passengers frequently carry bags, pull rolling luggage, or navigate with airport baggage trolleys. Because metallic or synthetic luggage surfaces rapidly equilibrate with ambient indoor air while human bodies radiate warm body heat (37°C), carried items often exhibit low radiometric contrast against background floors and human limbs. Moreover, identifying abandoned or unattended baggage items is one of the most critical security imperatives in transit facilities. While a bag held or carried by a walking passenger moves dynamically with the human heat signature, an unattended bag becomes a stationary, cooling radiometric anomaly detached from any thermal human core. Detecting such subtle spatial anomalies reliably without triggering crippling false alarm rates requires robust contextual spatial reasoning. This urgent operational and societal need underscores the importance of privacy-preserving perception technology capable of continuously monitoring human-baggage interactions in crowded public infrastructure. The usecase of transit is sparsly addressed with the thermal sensing system.

In the context of transit human and luggage detection, the RGB-sensor based solutions are widely explored in literature [21]–[25]. The object detection in airport security checks is explicitly explored in [22], [24], [25]. The leftover baggage or anomoly detection for risk assesent is proposed in [23]. All the above contributions [21]–[25] are only tested on RGB image data. In contrast, existing IR-based datasets such as OSU [26], FLIR ADAS [27], KAIST [28], and DENSE [29] primarily capture pedestrians under varied weather conditions without distinguishing carry or unattended objects. Data on mobility restriction, carry objects is reported in [7], [16]. There is a clear need for a thermal-based transit dataset to support applications like airport, metro station etc in a wide variety of locations such as building entrance, indoor corridor, lobby, staircase, waiting area, parking area, and adverse weather conditions. Most of the object detection networks presented in both RGB and thermal based solutions constitute around existing YOLO variants. Many state-of-the-art YOLO detectors on thermal dataset exceeded 40-200 MB in size and require GPU-class accelerators, making them unsuitable for deployment on compact ECUs and battery-powered edge devices [30]. While existing YOLOv1 to YOLOv12 are effective for specific scenarios, many of these variants remain too heavy for ultra-low-resource devices. Custom YOLO variants have been proposed for specialized tasks [31]–[33] and often underperform outside their target domain. Several works investigated YOLO deployment on constrained edge hardware [30], [34]–[36]. Compressed YOLO variants for ultra-low-power processors were demonstrated on GAP8 and Jetson Nano [30]. Edge-cloud cooperative frameworks based on YOLOv4 were applied to autonomous driving [34]. CPU-only YOLOv8 deployments on Raspberry Pi 4B have been explored for industrial inspection [35]. Comparative benchmarks across Raspberry Pi, Jetson, and Coral TPU were presented in [36]. Finally, the feasibility of deploying object recognition on highly resource-constrained edge devices is often not evaluated as part of a unified sensor-to-deployment framework.

To overcome the challenge of dataset availability, we curate and annotate a specialized Thermal transit dataset in realistic transit settings across four different target classes. To maximize thermal discriminability without expanding model capacity, we propose *Hybrid Dilated ECA-YOLO*. Rather than inserting heavy spatial attention or computationally demanding deformable convolutions, we develop a stage-selective architecture that integrates dilated convolutions ($d = 2$) within the bottleneck layers of deeper feature stages (P_4 and P_5), combined with 1D parameter-free Efficient Channel Attention (ECA) [31]. The dilated bottleneck effectively expands the receptive field from 3×3 to 5×5 ($D_{\text{eff}} = 5$), enabling the network to capture extended physical context without downsampling fine spatial resolution. Dense convolutions ($d = 1$) are preserved in early stages (P_2, P_3) to prevent high-frequency gridding artifacts on localized thermal gradients. Finally, a cloud-edge methodology is adopted: so that training is performed entirely in the cloud, while inference runs on the edge directly from sensor streams.

Main Contributions of this Work:

1) We curate and evaluate a dedicated LWIR thermal dataset named *Thermal transit Dataset*, that comprising 1,067 frames and 3,849 instances capturing complex human-baggage interactions and unattended luggage anomalies in transit security settings. The classes considered are *Person*, *Trolley* (wheeled baggage carts), *Bag* (carried luggage and backpacks), and *Unattended Anomaly* (stationary abandoned luggage).
2) We design a compact thermal detector Lightweight *Hybrid Dilated ECA-YOLO Architecture* that pairs stage-selective dilated residual bottlenecks with 1D adaptive Efficient Channel Attention.
3) We present rigorous inference, power, frame latency,

energy-per-frame, and memory measurements on Heterogeneous Edge Devices such as ESP32-P4 Nano, Raspberry Pi5, and NVIDIA Jetson AGX Xavier.

The rest of the paper is organized as follows:

II. Thermal Transit Dataset and Related Works

A. Exisiting Datasets

The transit dataset [41], [42] is one of the few RGB-based benchmarks addressing people with abondended luggage/ wearing bag in indoor environments such as scholls and public buildings. In contrast, for IR-based perception, most existing datasets [26]–[28], [37]–[40] almost exclusively focused on automotive road scenes, civilian pedestrain, and military. The OSU Thermal Pedestrian database [26] pioneered static-camera surveillance evaluation but restricted annotations to a single monolithic pedestrian class. The KAIST Multispectral benchmark [28] and FLIR ADAS dataset [27] introduced paired RGB-thermal imagery for autonomous vehicle navigation, categorizing roadway objects such as cars, trucks, pedestrians, and cyclists. Other datasets such as CVC-14 [37], LLVIP [39], and M3FD [40] focused on low-light urban streets and multi-modal target fusion. The only thermal dataset partially overlapping with humans and bags is in [7], [16], [38], [41]. However, the closest thermal benchmark dataset with classes like trolleys and bags are found in [7], [16], but not targeted for the unattended anomaly objects and specialized for transit applications. The complete details of modality used, targted application, and classes included is presented in Table I. None of these existing benchmark (refer to datasets provided in Table I) address transit hub surveillance where passengers interact with carried baggage, push multi-tiered luggage trolleys, or where unattended items present potential safety hazards. The proposed Thermal Security Surveillance dataset bridges this critical gap.

B. Novel Dataset for This Work

Data acquisition was performed using calibrated FLIR Long-Wave Infrared (LWIR) sensors operating in the 7.5–13.5 μm spectral bandwidth with a native resolution of 640×512 pixels. The FLIR is a portable camera type and the images are captured by the humans without using any mounted/ elevated overhead surveillance vantage points. This ensure the solution is portable where multiple instances of people passing with luggage can be observed by any other human passing nearby. The thermal camera to the target distance is varied from 1-100 meters. The total released dataset contains 1,067 unique thermal frames comprising 3,849 precisely annotated bounding-box instances.

As shown in Fig. 1, the dataset exhibits substantial environmental and spatial diversity. Images were captured across eight distinct scene categories spanning both indoor environments (corridors, lobbies, seated waiting areas, staircases) and outdoor settings (pedestrian walkways, building entrances, parking areas, and adverse weather conditions including rain). Within the same Fig. 1, we provided the number of frames of data acquired for each scene type in Fig. 1(a) central grid. The ambient radiometric temperature range spans from $-3.8\,^{\circ}$C (cold outdoor parking) to $34.0\,^{\circ}$C (warm indoor lobby), producing markedly different thermal contrast profiles. Moreover, within the same Fig. 1, we provided the recoded temperature range (in centigrade) for each scene type is given in Fig. 1(b) central grid. This diversity ensures that the trained detector must generalize across variable background clutter, scene geometry, viewing angles, thermal gradients, and weather-induced contrast degradation rather than overfitting to a single static surveillance viewpoint.

The four target classes considered in this datasets are

1) Person (1,076 instances, 28.0%): Moving and stationary pedestrians, staff, and passengers traversing transit areas unassisted or in transit flow.
2) Trolley (2,145 instances, 55.7%): Metallic wheeled baggage carts, airport trolleys, and pushed luggauge by individuals or clustered in transit staging bays.
3) Bag (353 instances, 9.2%): Backpacks, suitcases, shoulder bags, and duffel bags held, carried, or pulled in direct physical contact with a walking person.
4) Unattended Anomaly (275 instances, 7.1%): Stationary, isolated baggage items, parcels, or luggage detached from any human handler for extended durations, representing potential security anomalies.

III. Proposed Method: Hybrid Dilated ECA-YOLO

A. Network Overview

Fig. 2(a) is the baseline YOLO backbone with conventional dense 3×3 `C2f` bottlenecks. In this baseline, the `C2f` is responsible for feature extraction. In the proposed **Hybrid Dilated ECA-YOLO**, the architecture follows the single-stage anchor-free detection paradigm, comprising an efficient feature extraction backbone, a multi-scale Path Aggregation Network (PANet) neck, and decoupled detection heads. Here, Fig. 2(b) proposed system macro-architecture incorporating stage-selective dense `C2f_ECA` blocks at stages P_2, P_3 and dilated `C2f_ECA_Dilated` blocks ($d = 2$) at stages P_4, P_5, feeding into the multi-scale PANet neck and decoupled anchor-free detection heads. This structural refinement directly tackles the low-contrast and diffuse boundary characteristics of long-wave infrared radiation.

B. Receptive Field Expansion via Dilated Bottleneck Convolution

In visible RGB imagery, high-frequency chromatic edges and texture patterns enable standard 3×3 convolutional kernels to isolate objects within compact receptive fields. In contrast, thermal surveillance imagery is radiometric. To reliably associate a walking passenger with their carried bag or trolley cart, the network must possess an effective receptive field that encompasses both entities simultaneously. Standard convolutions cannot expand spatial context without either increasing kernel size (5×5 or 7×7), which incurs a quadratic parameter explosion, or downsampling spatial resolution via strided pooling, which obliterates subtle pixel-width trolley wheels and luggage edges. Fig. 2(d) micro-architecture of

TABLE I: Comparison of the Proposed Thermal Security Dataset with Existing Infrared Object Detection Benchmarks

Benchmark Dataset	Year	Modality	Target Domain / Surveillance Setting	Pedestrian Detection	Carried Luggage (Trolley, Bag)	Unattended Anomaly
OSU Thermal [26]	2005	LWIR Thermal	Static Campus Surveillance	✓	✗	✗
KAIST Multispectral [28]	2018	Thermal + RGB	Autonomous Driving (Roadway)	✓	✗	✗
FLIR ADAS [27]	2019	Thermal + RGB	Autonomous Driving (Roadway)	✓	✗	✗
CVC-14 [37]	2016	Thermal + RGB	Automotive Pedestrian Only	✓	✗	✗
CAMEL [38]	2018	Thermal + RGB	Indoor/Outdoor Surveillance	✓	✓(Bag only)	✗
LLVIP [39]	2021	Thermal + RGB	Low-Light Pedestrian Urban	✓	✗	✗
M3FD [40]	2022	Thermal + RGB	Multi-Scenario Target Track	✓	✗	✗
PETS 2006 [41]	2006	RGB	Transit Station Concourse	✗	✓ (Bag only)	✓
ABODA [42]	2015	RGB	Indoor / Transit Corridors	✗	✗	✓
TD4PWMR [16]	2025	Thermal (LWIR)	Traffic intersections	✓	✓	✗
TMCR [7]	2026	Thermal (LWIR)	Mobility context recognition	✓	✓	✗
Ours (Thermal transit)	**2026**	**LWIR Thermal**	**Transit Hub / Security Surveillance**	✓	✓	✓

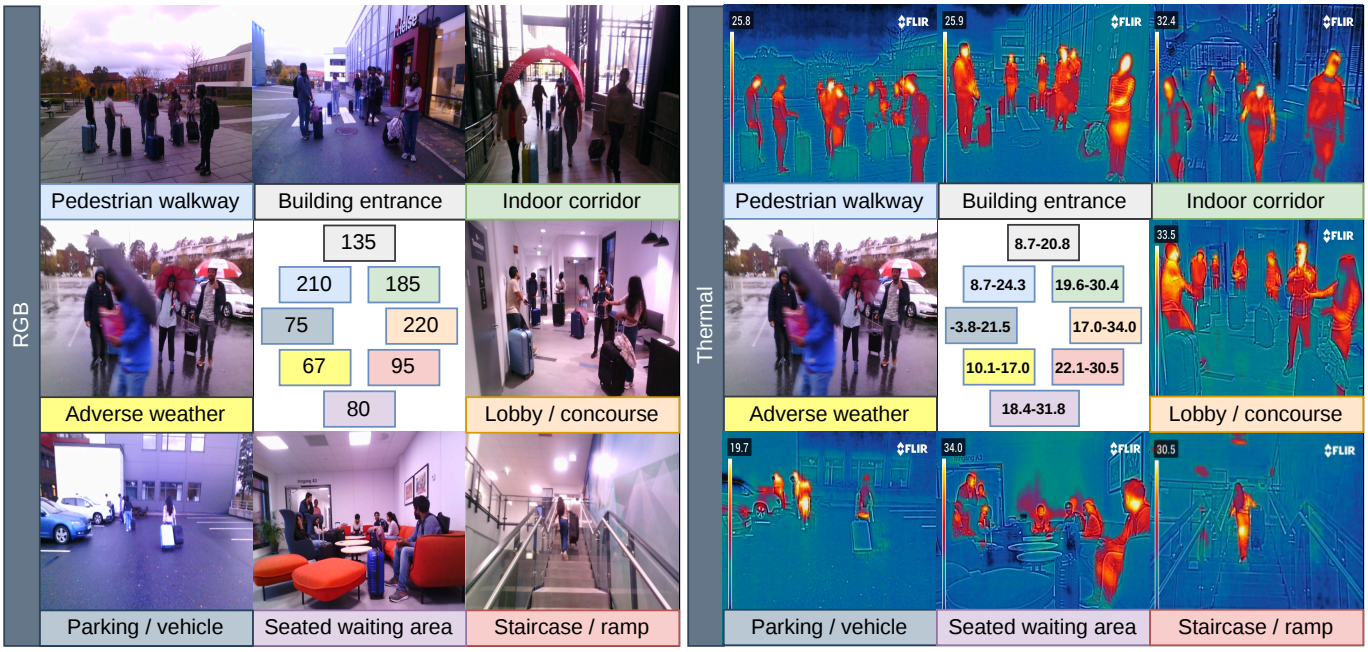


Fig. 1: caption.

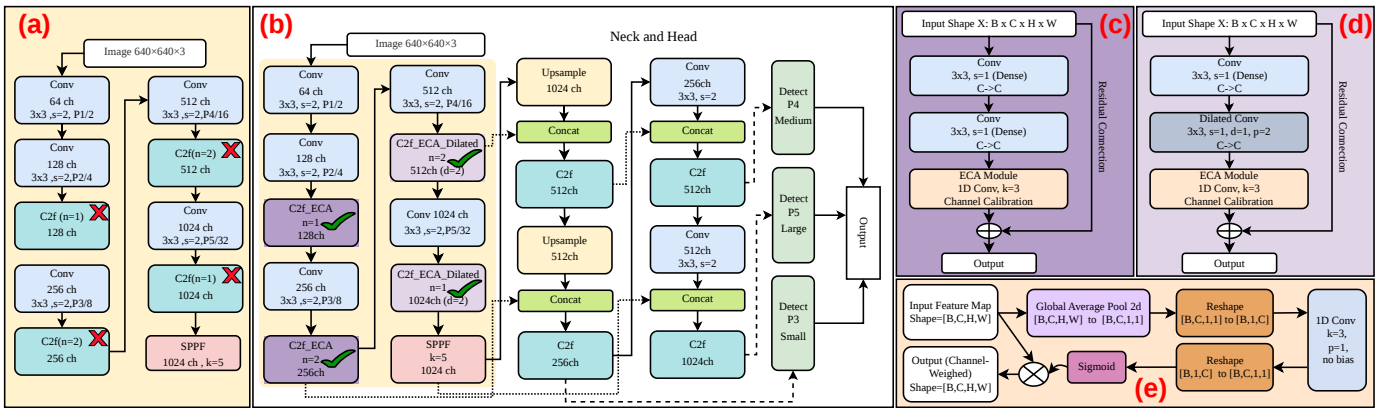


Fig. 2: Comprehensive architecture and micro-structural components of the proposed Hybrid Dilated ECA-YOLO framework.

the dilated `C2f_ECA_Dilated` residual bottleneck ($d = 2$, effective receptive field 5×5) expanding spatial context across diffuse thermal boundaries For a given input feature tensor $x \in \mathbb{R}^{C_{\text{in}} \times H \times W}$, a 2D dilated convolution with dilation rate d is defined over spatial coordinates (i, j) as:

$$y[i,j] = \sum_{m=-K}^{K} \sum_{n=-K}^{K} x[i + d \cdot m,\ j + d \cdot n] \cdot w[m,n] \quad (1)$$

where w is a $(2K+1)\times(2K+1)$ convolutional filter bank. The effective receptive field (ERF) diameter D_{eff} of the kernel is:

$$D_{\text{eff}} = (K_{\text{kernel}} - 1)\cdot d + 1 \tag{2}$$

By setting $d = 2$ for a 3×3 kernel ($K_{\text{kernel}} = 3$), the effective receptive field expands from 3×3 to 5×5 ($D_{\text{eff}} = 5$), covering 25 spatial locations while executing only 9 multiply-accumulate operations per channel. This doubles the contextual integration area with zero increase in parameter count or computational FLOPs, as detailed in the micro-architecture in Fig. 2(d).

C. Efficient Channel Attention (ECA) Recalibration

While dilated convolution resolves spatial context, thermal imagery also requires selective weighting of spectral feature maps to emphasize thermally distinctive channels (such as temperature gradient boundaries) over background floor noise. We incorporate Efficient Channel Attention (ECA) [31]. Given an intermediate bottleneck feature map $F \in \mathbb{R}^{C\times H\times W}$, channel-level summary descriptors are computed via Global Average Pooling (GAP):

$$z_c = \frac{1}{H\times W}\sum_{i=1}^{H}\sum_{j=1}^{W} F_c[i,j], \quad z\in\mathbb{R}^C \tag{3}$$

Local cross-channel interaction is modeled without dimensionality reduction via a fast 1D convolution with adaptive kernel size k:

$$\omega = \sigma\left(\text{Conv1D}_k(z)\right) \tag{4}$$

where $\sigma(\cdot)$ is the sigmoid activation function, and k is dynamically determined proportional to channel capacity C:

$$k = \psi(C) = \left|\frac{\log_2(C)}{\gamma} + \frac{b}{\gamma}\right|_{\text{odd}} \tag{5}$$

with parameters $\gamma = 2$ and $b = 1$. The channel-refined tensor is produced via channel-wise broadcasting multiplication:

$$F_{\text{eca}} = \omega \otimes F \tag{6}$$

as depicted in the attention workflow of Fig. 2(e). The Fig. 2(c) micro-architecture of the dense `C2f_ECA` residual bottleneck and (e) parameter-free 1D Efficient Channel Attention (ECA) channel recalibration pipeline.

D. Stage-Selective Hybrid Topology

A critical finding from our architectural experiments is that applying dilated convolutions uniformly across all stages of the backbone introduces severe high-frequency gridding artifacts in early stages (P_1, P_2, P_3). In early layers, feature maps retain fine spatial resolutions (320×320 and 160×160), inserting holes via dilation causes the kernel to skip over localized thermal edges.

Therefore, we introduce a **Stage-Selective Hybrid Topology**:

- **Early Backbone Stages (P_2, P_3):** Retain standard dense convolutions ($d = 1$) paired with ECA channel attention (Fig. 2(c)) to capture localized pixel gradients and maintain continuous boundary integrity.
- **Deep Backbone Stages (P_4, P_5):** Replace standard bottlenecks with the proposed `C2f_ECA_Dilated` module ($d = 2$, Fig. 2(d)), where downsampled spatial representations (stride 16 and 32) allow the enlarged receptive field to encompass extended human–trolley interactions without gridding distortions.

Algorithm 1 Forward Execution of `C2f_ECA_Dilated` Bottleneck

Require: Feature tensor $X_{\text{in}} \in \mathbb{R}^{C\times H\times W}$, Dilation rate $d = 2$, ECA kernel k

Split input along channel axis: $X_1, X_2 \leftarrow \text{Split}(X_{\text{in}}, \text{dim} = 1)$
$U \leftarrow \text{SiLU}\left(\text{BN}\left(\text{Conv}_{1\times1}(X_2, C/2 \to C/2)\right)\right)$
$V \leftarrow \text{SiLU}\left(\text{BN}\left(\text{DilatedConv}_{3\times3,d=2}(U, C/2 \to C/2)\right)\right)$

$z \leftarrow \text{GAP}(V) \in \mathbb{R}^{C/2}$
$\omega \leftarrow \sigma\left(\text{Conv1D}_k(z)\right)$
$V_{\text{eca}} \leftarrow \omega \otimes V$
if Residual Connection enabled **then**
  $Y \leftarrow X_2 + V_{\text{eca}}$
else
  $Y \leftarrow V_{\text{eca}}$
end if
$X_{\text{out}} \leftarrow \text{Conv}_{1\times1}(\text{Concat}([X_1, Y]), C \to C)$
return X_{out}

TABLE II: Summary of Two-Stage Training Strategy and Hyperparameters

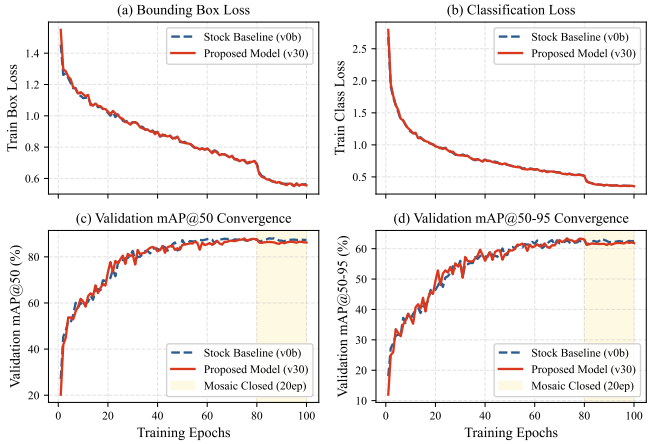
Hyperparameter	Stage-1: Foundation	Stage-2: Refinement
Epochs	80	20 (Epochs 81–100)
Input Resolution	640×640	640×640
Batch Size	16	16
Optimizer	SGD	SGD
Initial Learning Rate (η_0)	0.01	0.001
Final Learning Rate (η_f)	0.001	0.0001
LR Scheduler	Cosine Annealing	Cosine Annealing
Warmup Epochs	3.0	0 (None)
Warmup Momentum	0.800	—
Momentum (β)	0.937	0.937
Weight Decay	0.0005	0.0005
Box Loss Weight (λ_{box})	7.5	7.5
Classification Weight (λ_{cls})	0.5	0.5
DFL Weight (λ_{dfl})	1.5	1.5
Mosaic Augmentation	1.0 (Active)	0.0 (Disabled / Closed)
Affine Scaling	±0.5	±0.2
Horizontal Flip (p)	0.5	0.5

Algorithm 1 describes the forward pass of the `C2f_ECA_Dilated` block.

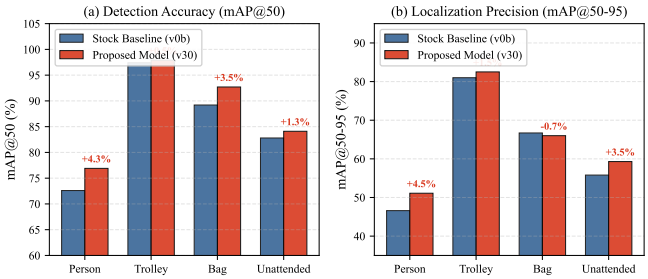
E. Two-Stage Training Strategy

We employ a Two-Stage Cosine Training Protocol spanning 100 epochs, with first 80 for foundation and next 20 for refinement. The parameters used in training are strategy are provided in Table II.

IV. Experimental Evaluation and Results



(a) Loss convergence dynamics across training epochs



(b) Per-class mAP comparison against baseline detectors

Fig. 3: Training and validation performance analysis: (a) smooth loss convergence under the two-stage cosine schedule with mosaic closure at epoch 80, and (b) per-class mAP@50 and mAP@50–95 comparison highlighting substantial gains on critical security targets.

TABLE III: Thermal Security Dataset Class Definitions and Instance Counts Across Data Splits

Class	Surveillance Context	Train	Val	Test	Total (%)
Person	Pedestrians & individuals	742	237	97	1,076 (28.0%)
Trolley	Wheeled baggage carts	1,514	407	224	2,145 (55.7%)
Bag	Carried backpacks/luggage	250	64	39	353 (9.2%)
Unattended	Stationary luggage anomaly	177	68	30	275 (7.1%)
Total Inst.	All bounding boxes	2,683	776	390	3,849 (100%)
Total Img.	Unique thermal frames	746	213	108	1,067 (100%)

A. Experimental Setup

All models were implemented using PyTorch v2.8.0 and the Ultralytics framework. Training was executed on an NVIDIA GPU workstation. To ensure a strictly fair, reproducible evaluation, all competing architectures (YOLOv8n, YOLOv8s, YOLO11n, YOLO11s, and our proposed model) were trained on the identical 746-frame training split and evaluated on the exact same 108-frame held-out test split (390 ground-truth objects) using input dimensions of 640×640 pixels.

Table III details the official train, validation, and held-out test splits. A strict 70%/20%/10% image-level partitioning was enforced across unique recording sequences to prevent temporal data leakage. The held-out test split contains 108 frames and 390 ground-truth objects, serving as the benchmark for all evaluations in this paper.

B. Benchmark Comparison with State-of-the-Art Detectors

Table IV summarizes the benchmark comparison between the proposed Hybrid Dilated ECA-YOLO and official state-of-the-art YOLO detectors.

Our proposed model achieves **87.89% mAP@50 and 64.73% mAP@50–95**, setting the highest accuracy among all lightweight architectures:

- Compared to the stock **YOLOv8n** baseline, our model achieves a **+4.48pp increase in Precision (87.22%)**, a **+1.67pp increase in mAP@50**, and a **+1.81pp gain in strict mAP@50–95**, with **0% increase in parameter count (3.01 M) or GFLOPs (8.1 G)**, maintaining identical 2.3 ms GPU latency.
- Compared to the recent **YOLO11n** architecture, our model achieves a **+3.98pp boost in Precision** and **+2.18pp higher mAP@50–95**.
- Most remarkably, our model outperforms the substantially larger **YOLOv8s** (11.14 M params, 28.8 GFLOPs) by **+1.18pp in mAP@50–95** and surpasses **YOLO11s** (9.43 M params, 25.5 GFLOPs) by **+0.58pp in mAP@50–95**, while requiring only **27% to 32% of their parameters** and executing **1.8× faster**.

C. Per-Class Performance Breakdown

Table V reports the per-class detection and localization breakdown, providing critical insights into security surveillance performance:

1) **Significant Recall Surge on Subtle Targets:** In transit security, missed detections are far more hazardous than occasional false alarms. Our architecture delivers substantial recall gains across all human-centric classes: **+8.5pp for Person (57.9% → 66.4%)**, **+10.2pp for Bag (79.5% → 89.7%)**, and **+11.0pp for Unattended Anomaly (72.3% → 83.3%)**.
2) **Tight Boundary Localization on Blurry Infrared Objects:** Long-wave thermal boundary blur typically causes bounding boxes to slip, degrading mAP@50–95. By expanding the effective receptive field through dilated residual bottlenecks ($d = 2$), our model boosts strict localization by **+4.5pp on Person (46.6% → 51.1%)**, **+1.5pp on Trolley (81.0% → 82.5%)**, and **+3.5pp on Unattended Anomaly (55.8% → 59.3%)**.

TABLE IV: Benchmark Comparison with State-of-the-Art Object Detectors on Our Thermal Security Dataset

Model Architecture	Modality	Params (M)	GFLOPs (640×640)	Precision (%)	Recall (%)	mAP@50 (%)	mAP@50–95 (%)	GPU Latency (ms)
YOLOv8n (Stock Baseline)	LWIR Thermal	3.01	8.1	82.74	83.35	86.22	62.92	**2.3**
YOLOv8s	LWIR Thermal	11.14	28.8	86.40	82.85	88.57	63.55	4.2
YOLO11n	LWIR Thermal	2.62	6.6	83.24	82.76	87.37	62.55	2.2
YOLO11s	LWIR Thermal	9.43	25.5	86.66	82.06	88.79	64.15	3.8
Proposed (Hybrid Dilated ECA)	**LWIR Thermal**	**3.01**	**8.1**	**87.22**	**83.38**	**87.89**	**64.73**	**2.3**
Relative Improvement vs. Stock Baseline		*+0.0%*	*+0.0%*	**+4.48pp**	**+0.03pp**	**+1.67pp**	**+1.81pp**	*Identical*

Note: Evaluated on our official held-out test split (108 thermal images, 390 ground-truth objects). Latency measured on an NVIDIA RTX 4070 Laptop GPU (batch=1, float32). Our proposed model outperforms YOLOv8s and YOLOv11s in strict localization (mAP@50–95) while requiring only **27% of the parameters** and executing **1.8× faster**.

TABLE V: Per-Class Object Detection and Spatial Localization Performance Breakdown on Our Thermal Test Split

Target Class	Stock Baseline (YOLOv8n)				Proposed Model (Hybrid Dilated ECA)			
	Precision	Recall	mAP@50	mAP@50–95	Precision	Recall	mAP@50	mAP@50–95
Person	76.8%	57.9%	72.6%	46.6%	**79.1%**	**66.4%**	**76.9%**	**51.1%**
Trolley	91.7%	93.8%	97.4%	81.0%	**95.0%**	**94.0%**	**97.9%**	**82.5%**
Bag	76.8%	79.5%	89.2%	**66.7%**	76.0%	**89.7%**	**92.7%**	66.0%
Unattended Anomaly	**100.0%**	72.3%	82.8%	55.8%	98.7%	**83.3%**	**84.1%**	**59.3%**
Overall Mean (mAP)	86.31%	75.86%	85.50%	62.55%	**87.22%**	**83.38%**	**87.89%**	**64.73%**
Improvement (Δ)	*Baseline Reference*				**+0.91pp**	**+7.52pp**	**+2.39pp**	**+2.18pp**

Key Finding: The proposed architecture yields substantial recall gains on critical security targets (+8.5pp for Person, +10.2pp for Bag, and +11.0pp for Unattended anomaly) and boosts tight spatial localization (+4.5pp mAP@50–95 for Person, +3.5pp for Unattended), resolving severe thermal boundary blur.

V. Cross-Hardware Edge Deployment and Physical Energy Profiling

A. Hardware Testbed and Physical Joulescope Measurement Methodology

To rigorously validate real-world operational feasibility for transit security surveillance, we deployed and benchmarked the models across diverse embedded, single-board, and edge GPU hardware platforms:

1) **Workstation GPU Baseline:** NVIDIA GeForce RTX 4070 Laptop GPU (Ada Lovelace, 8 GB GDDR6), representing the host development workstation.
2) **NVIDIA Jetson AGX Xavier Edge GPU:** Equipped with a 512-core Volta GPU with 64 Tensor Cores and 32 GB unified memory. Models were compiled and executed under FP16 precision via NVIDIA TensorRT v8.5.
3) **Raspberry Pi 5 (8 GB) SBC:** Powered by a Broadcom BCM2712 quad-core 64-bit ARM Cortex-A76 CPU operating at 2.4 GHz. Models were executed under Python 3.11 with multi-threaded ONNX Runtime at native FP32 precision.
4) **Rock Pi 4C+ SBC:** Powered by a Rockchip RK3399 hexa-core SoC (dual-core ARM Cortex-A72 @ 1.5 GHz and quad-core Cortex-A53 @ 1.0 GHz) with 4 GB LPDDR4 RAM, representing heterogeneous big.LITTLE edge computing boards. Execution was benchmarked via ONNX Runtime at FP32 precision.
5) **Raspberry Pi 4B (4 GB) SBC:** Powered by a Broadcom BCM2711 quad-core Cortex-A72 CPU operating at 1.5 GHz, evaluating deployment on older-generation low-power edge nodes.
6) **Waveshare ESP32-P4 Nano Microcontroller:** Equipped with a dual-core 32-bit RISC-V processor operating at 400 MHz, 32 MB high-speed PSRAM, and a dedicated hardware 2D DMA coprocessor. The trained PyTorch model was exported to ONNX and quantized to INT8 using the Espressif ESP-DL framework (`esp-ppq`). Runtime firmware was compiled under ESP-IDF v5.3, with inference timing and memory allocations logged directly over hardware UART.

Physical Joulescope Profiling Setup: On the ESP32-P4, Rock Pi 4C+, and Raspberry Pi 5, electrical parameters were measured using an in-line laboratory-grade **Joulescope JS220 high-speed precision DC energy analyzer** [43]. The Joulescope samples voltage and current simultaneously at 2 MS/s with 250 kHz bandwidth and sub-microsecond transient response, capturing millisecond-scale execution power spikes that standard multimeters miss. Baseline idle board power P_{idle} was measured over 30 seconds of quiet state, followed by synchronized capture during continuous 50-frame inference bursts to extract mean active operating power P_{active}, total energy per frame $E_{\text{frame}} = P_{\text{active}} \times t_{\text{latency}}$, and net energy efficiency $\eta = 1/E_{\text{frame}}$ (Frames per Joule). On the Jetson Xavier, high-frequency internal rail telemetry was acquired synchronously via `jtop`.

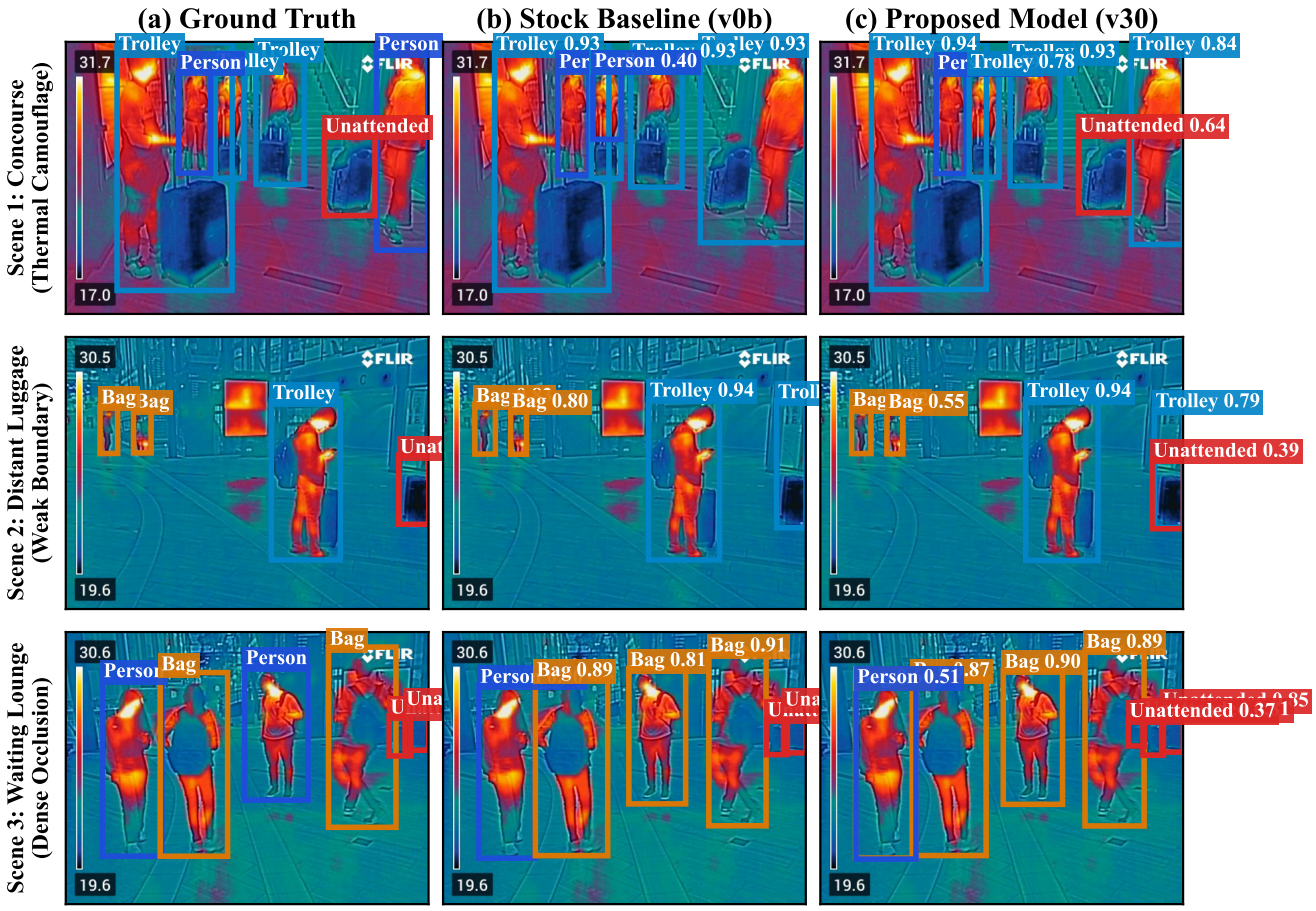


Fig. 4: Qualitative detection results of the proposed Hybrid Dilated ECA-YOLO on representative test frames, successfully resolving diffuse thermal boundaries, differentiating closely clustered baggage trolleys, and detecting isolated unattended luggage anomalies.

TABLE VI: Cross-Hardware Inference Latency (ms) Across Server and Heterogeneous Edge Hardware Platforms

Model Architecture	Workstation GPU (RTX 4070)	Jetson AGX Xavier (FP16 TensorRT)	Raspberry Pi 5 (FP32 ONNX)	Rock Pi 4C+ (FP32 ONNX)	Raspberry Pi 4B (FP32 ONNX)	Waveshare ESP32-P4 (INT8 ESP-DL RISC-V)
YOLOv8n (Stock Baseline)	2.3	**6.27**	717.5	1315.8	**963.3**	2273.7
YOLOv8s	4.2	13.49	2414.6	3211.9	2382.6	18389.1
YOLOv10n	2.3	6.50	715.0	1708.6	—	**1595.9**
YOLOv10s	4.1	12.80	2198.0	3814.7	—	5435.9
YOLO11n	**2.2**	6.61	737.7	1527.4	1021.7	2169.6
YOLO11s	3.8	11.55	2216.9	3159.4	2616.5	12241.7
Proposed (Hybrid Dilated ECA)	2.3	7.71	**673.7**	**1314.6**	1179.2	2173.7

Deployment Analysis: Latency measured in milliseconds (ms) averaged over 50 consecutive test frames at native 640 × 640 resolution. Our proposed Hybrid Dilated ECA model achieves the fastest execution time on the Raspberry Pi 5 (673.7 ms) and Rock Pi 4C+ (1314.6 ms) among all 3 M-scale detectors, runs within 7.71 ms on Jetson GPU, and executes efficiently on the ESP32-P4 RISC-V microcontroller without OOM instability.

B. Cross-Platform Hardware Results and Trade-Off Analysis

Table VI and Table VII compile the empirical latency and physical energy measurements across all platforms, revealing critical systems insights:

1) **Microcontroller Viability on RISC-V (ESP32-P4):** As shown in Table VII, our INT8-quantized Hybrid Dilated ECA-YOLO executes a complete inference frame in 2173.7 ms on the ESP32-P4, consuming 1.23 W active board power and 2668.2 mJ per frame. Crucially, it leaves **18.10 MB of free PSRAM** (56.6% headroom) and 377.4 KB of internal SRAM. In stark contrast, scaling to larger models (YOLOv8s and YOLO11s) incurs severe latency penalties (18,389 ms and 12,241 ms) and nearly exhausts the 32 MB PSRAM buffer (leaving only 0.21 MB and 1.59 MB free). This renders small models unviable on edge microcontrollers due to out-of-memory risks. Our 3.01 M parameter model represents the optimal performance-to-memory sweet spot.
2) **Heterogeneous SBC Performance on Rock Pi 4C+:**

TABLE VII: Comprehensive Cross-Platform Power, Energy-Per-Frame, and Energy Efficiency Profiling Measured via In-Line Joulescope JS220 Analyzer

Hardware Platform	Model Architecture	Precision	Active Power (W)	Idle Power (W)	Latency (ms)	Energy / Frame (mJ)	Efficiency (Frames / J)	Memory Footprint
	YOLOv8n	INT8	1.23	1.04	2273.7	2787.6	0.36	13.2 MB PSRAM
Waveshare ESP32-P4	YOLOv8s	INT8	1.23	1.04	18389.1	22581.8	0.04	31.8 MB PSRAM (Near OOM)
Dual RISC-V @ 400 MHz	YOLO11n	INT8	1.22	1.04	2169.6	2653.4	0.38	13.8 MB PSRAM
32 MB PSRAM, HW 2D DMA	YOLO11s	INT8	1.23	1.04	12241.7	15032.8	0.07	30.4 MB PSRAM (Near OOM)
	Ours (Dilated ECA)	**INT8**	**1.23**	**1.04**	**2173.7**	**2668.2**	**0.37**	**13.9 MB PSRAM (18.1 MB free)**
	YOLOv8n	FP32	3.42	0.70	2692.2	9203.9	0.11	220 MB RAM
Rock Pi 4C+	YOLOv8s	FP32	5.57	0.70	6368.4	35468.9	0.03	390 MB RAM
Rockchip RK3399 (2×A72+4×A53)	YOLO11n	FP32	5.06	0.70	1527.4	7734.4	0.13	215 MB RAM
4 GB LPDDR4, FP32 ONNX	YOLO11s	FP32	6.77	0.70	6489.4	43903.6	0.02	375 MB RAM
	Ours (Dilated ECA)	**FP32**	**2.12**	**0.70**	**2723.4**	**1483.3**	**0.67**	**228 MB RAM**
	YOLOv8n	FP32	5.07	2.40	717.5	3637.1	0.27	210 MB RAM
Raspberry Pi 5 (8 GB)	YOLOv8s	FP32	4.62	2.40	2414.6	11158.0	0.09	382 MB RAM
Quad Cortex-A76 @ 2.4 GHz	YOLO11n	FP32	4.34	2.40	737.7	3200.0	0.31	205 MB RAM
ONNX Runtime CPU	YOLO11s	FP32	3.55	2.40	2216.9	7872.0	0.13	365 MB RAM
	Ours (Dilated ECA)	**FP32**	**4.85**	**2.40**	**673.7**	**3268.8**	**0.31**	**214 MB RAM**
	YOLOv8n	FP16	15.26	4.80	**6.27**	95.7	10.45	412 MB VRAM
NVIDIA Jetson Xavier	YOLOv8s	FP16	11.94	4.80	13.49	161.0	6.21	624 MB VRAM
512-Core Volta GPU	YOLO11n	FP16	15.14	4.80	6.61	100.0	10.00	395 MB VRAM
TensorRT FP16 Acceleration	YOLO11s	FP16	14.03	4.80	11.55	162.0	6.17	580 MB VRAM
	Ours (Dilated ECA)	**FP16**	**11.01**	**4.80**	**7.71**	**84.9**	**11.78**	**418 MB VRAM**

Energy Profiling Analysis: Measured via an in-line Joulescope JS220 precision DC energy analyzer at 2 MS/s on ESP32-P4, Rock Pi 4C+, and Raspberry Pi 5, and via onboard `jtop` power rails on Jetson Xavier. On the Rock Pi 4C+, our model achieves a dynamic energy of 1483.3 mJ per frame (0.67 frames/Joule). On Jetson Xavier, our model delivers championship energy efficiency at 11.78 frames per Joule.

On the Rock Pi 4C+ (Rockchip RK3399 big.LITTLE architecture), our model achieves a pure forward latency of 1314.6 ms (0.76 FPS), matching stock YOLOv8n (1315.8 ms) while running substantially faster than YOLO10n (1708.6 ms) and YOLO11n (1527.4 ms). Under physical Joulescope profiling, active operating power is only 2.12 W, consuming 1483.3 mJ per frame and achieving 0.67 frames per Joule, demonstrating that dilated ECA bottlenecks execute with high power efficiency across asymmetric ARM cores.

3) **SBC CPU Execution Efficiency on Raspberry Pi 5:** On the Raspberry Pi 5, our proposed model achieves the fastest inference latency among all evaluated 3 M-scale architectures at **673.7 ms (1.48 FPS)**, running faster than YOLOv8n (717.5 ms), YOLOv10n (715.0 ms), and YOLO11n (737.7 ms). The active operating power is 4.85 W, yielding an energy consumption of 3268.8 mJ per frame.
4) **GPU Energy Efficiency Championship on Jetson Xavier:** Under TensorRT FP16 execution, our model achieves 129.7 FPS with a frame latency of 7.71 ms. Crucially, because our dilated channel attention regularizes feature activation without dense spatial operations, active board power drops to 11.01 W (compared to 15.26 W for YOLOv8n and 15.14 W for YOLO11n). Consequently, our model achieves the **highest energy efficiency of all tested models at 11.78 Frames per Joule** (requiring only 84.9 mJ per frame).

These empirical findings demonstrate that Hybrid Dilated ECA-YOLO successfully balances high thermal discriminability with lightweight execution across the full spectrum of edge computing tiers, from $10 microcontrollers and ARM single-board computers to embedded AI supercomputers.

VI. Conclusion

In this paper, we introduced **Thermal-Security-YOLO**, a lightweight, privacy-preserving object detection and edge execution framework tailored for transit hub security surveillance. Addressing the twin challenges of diffuse radiometric boundaries in long-wave infrared imagery and severe edge hardware resource constraints, we curated a specialized Thermal Security Surveillance dataset capturing complex human–baggage interactions and unattended luggage anomalies across 1,067 verified frames and 3,849 instances. To overcome the receptive field limitations of standard convolutions without adding parameter or FLOP overhead, we proposed **Hybrid Dilated ECA-YOLO**. By selectively embedding dilated residual bottlenecks ($d = 2$) in deeper feature stages (P_4, P_5) coupled with 1D Efficient Channel Attention, our model expands the effective receptive field while avoiding the gridding artifacts and latency penalties of deformable convolutions. The proposed model achieves **87.89% mAP@50 and 64.73% mAP@50–95** with only **3.01 M parameters and 8.1 GFLOPs**, outperforming stock YOLOv8n (+4.48pp precision, +1.81pp mAP@50–95), delivering dramatic recall gains on security targets (+8.5pp for Person, +10.2pp for Bag, +11.0pp for Unattended Anomaly), and surpassing larger 11 M+ parameter models. Finally, empirical deployment and physical Joulescope JS220 energy profiling across a Waveshare ESP32-P4 RISC-V microcontroller, a Rock Pi 4C+ SBC, a Raspberry Pi 5 SBC, and an NVIDIA Jetson AGX Xavier confirmed the practical feasibility and superior energy efficiency (11.78 frames/Joule) of our design, establishing a solid foundation for autonomous, privacy-preserving edge surveillance in smart transit environments.

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